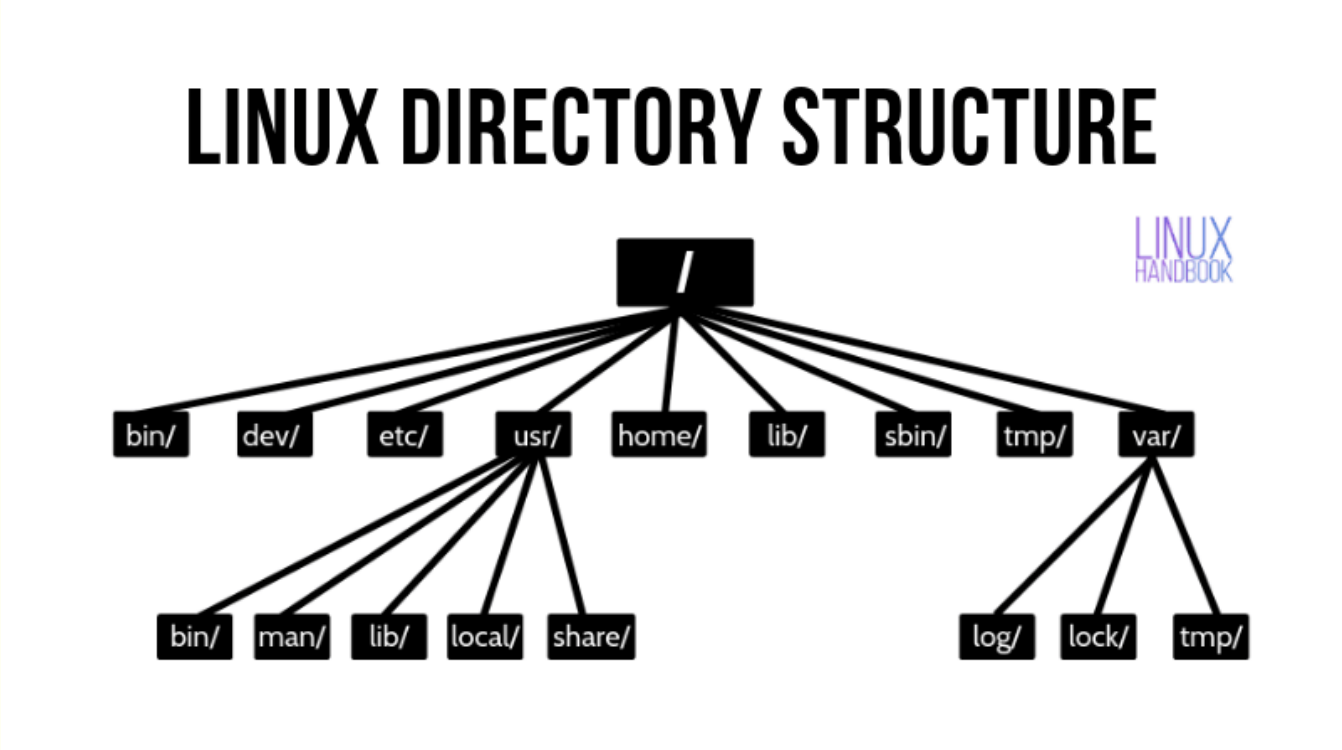
LINUX ESSENTIALS
CHAPTER 3
WORKING WITH FILES AND DIRECTORIES

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Objective

- In this lab, you will perform the following tasks:
  - List files and directories
  - Copy, move and delete files and directories



FILES AND DIRECTORIES

- On a Linux OS, data is stored in *files* and files are stored in *directories*. You may be used to the term *folders* to describe directories.

- Directories are actually files, too; the data that they hold are the names of the files that have been entered into the them, along with the inode number (a unique identifier number assigned to each file) for where the data for that file exists on the disk.

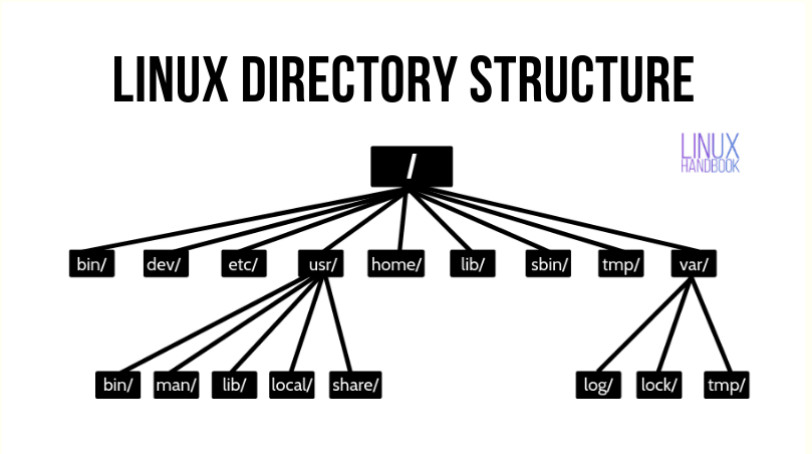
- As a Linux user, you will want to know how to manipulate these files and directories, including how to list files in a directory, copy, delete and move files.

Files & Directories

- Note:
  - File and directory names in Linux are case sensitive.
- command pwd used to print the *working directory.*
- The output of the pwd command (/home/sysadmin in the example above) is called the path. The first slash represents the root directory, the top level of the directory structure.
- When you first open a terminal window, you will be placed in your home directory.

```
sysadmin@localhost:~$ pwd
/home/sysadmin
sysadmin@localhost:~$
```

```
sysadmin@localhost:~$ echo $HOME
/home/sysadmin
sysadmin@localhost:~$
```



Files & Directories

- command cd with a path to a directory to change your current directory.
- To change back to your home directory, the cd command can be executed without a path.
- Note:
  - The tilde ~ character represents your home directory.

```
sysadmin@localhost:~$ cd /
sysadmin@localhost:/$ pwd
/
sysadmin@localhost:/$
```

```
sysadmin@localhost:/$ cd
sysadmin@localhost:~$ pwd
/home/sysadmin
sysadmin@localhost:~$
```

Files & Directories

- command cd with a path to a directory to change your current directory.
- To change back to your home directory, the cd command can be executed without a path.
- Note:
  - The tilde ~ character represents your home directory.
- Absolute paths are always complete paths from the root directory to a sub-directory or file.
- Relative pathnames you provide "directions" of where you want to go from the current directory.
- In relative path to change to the directory above the current directory (..)
- Change back to your home directory cd ~

```
sysadmin@localhost:~$ cd /
sysadmin@localhost:/$ pwd
/
sysadmin@localhost:/$
```

```
sysadmin@localhost:/$ cd
sysadmin@localhost:~$ pwd
/home/sysadmin
sysadmin@localhost:~$
```

```
sysadmin@localhost:/home$ cd ~
sysadmin@localhost:~$ pwd
/home/sysadmin
sysadmin@localhost:~$
```

```
sysadmin@localhost:/usr/share/doc/bash$ cd ..
sysadmin@localhost:/usr/share/doc$ pwd
/usr/share/doc
sysadmin@localhost:/usr/share/doc$
```

Files & Directories

- Command ls used to list the contents of the current directory.
- The color indicates what type the item is.
- Command ls -a used to display all files, including hidden files, use the -a option to the command.
  - Hidden files begin with a period (a dot character).
  - the .bashrc file shown in the example above contains configuration information for the bash shell.

```
sysadmin@localhost:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos
sysadmin@localhost:~$
```

Color	Type of File
Black or White	Regular file
Blue	Directory file
Cyan	Symbolic link file (a file that points to another file)
Green	Executable file (AKA, a program)

```
sysadmin@localhost:~$ ls -a
.             .bashrc   .selected_editor  Downloads  Public
..            .cache    Desktop           Music      Templates
.bash_logout  .profile  Documents         Pictures   Videos
sysadmin@localhost:~$
```

Files & Directories

- Command ls –l used to provide more information about a file.
- Command ls –R used to show contents of the subdirectories.
- Be careful of the -R option. Some directories are very, very large!

```
sysadmin@localhost:~$ ls -l /etc/hosts
-rw-r--r-- 1 root root 150 Jan 22 15:18 /etc/hosts
sysadmin@localhost:~$
```

```
sysadmin@localhost:~$ ls -R /etc/udev
/etc/udev:
rules.d  udev.conf

/etc/udev/rules.d:
70-persistent-cd.rules  README
sysadmin@localhost:~$
```

Files & Directories

- You can use file *globbing* (wildcards) to limit which files or directories you see.
- Note:
  - the -d option prevents files from subdirectories from being displayed.
  - It should always be used with the ls command when you are using file globbing.

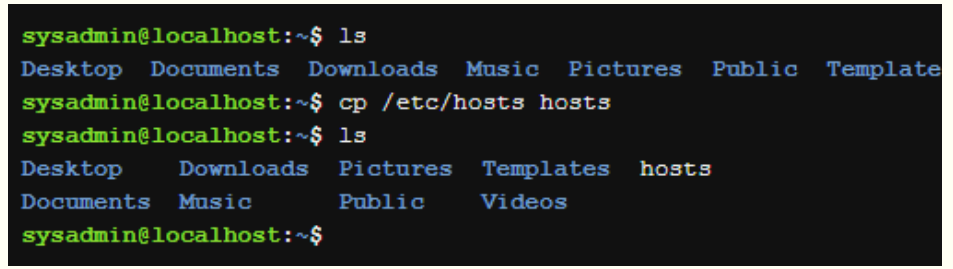
```
sysadmin@localhost:~$ ls -d /etc/s*
/etc/securetty  /etc/sgml     /etc/shells  /etc/ssl        /etc/sysctl.conf
/etc/security   /etc/shadow   /etc/skel    /etc/sudoers    /etc/sysctl.d
/etc/services   /etc/shadow-  /etc/ssh     /etc/sudoers.d  /etc/systemd
sysadmin@localhost:~$
```

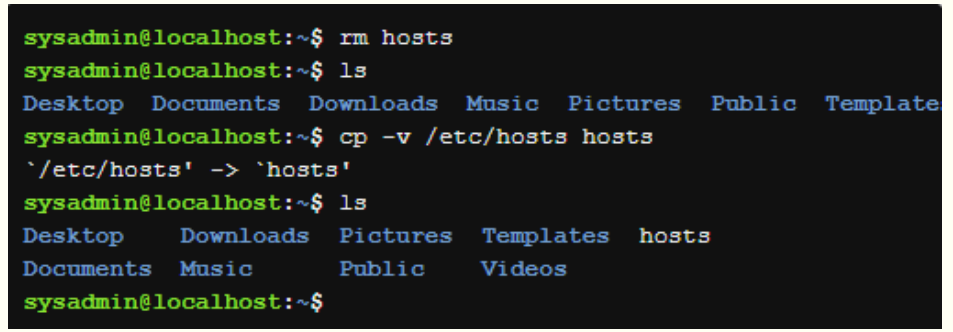
```
sysadmin@localhost:~$ ls -d /etc/????
/etc/bind  /etc/init  /etc/motd  /etc/perl  /etc/skel
/etc/dpkg  /etc/ldap  /etc/mtab  /etc/sgml  /etc/udev
sysadmin@localhost:~$
```

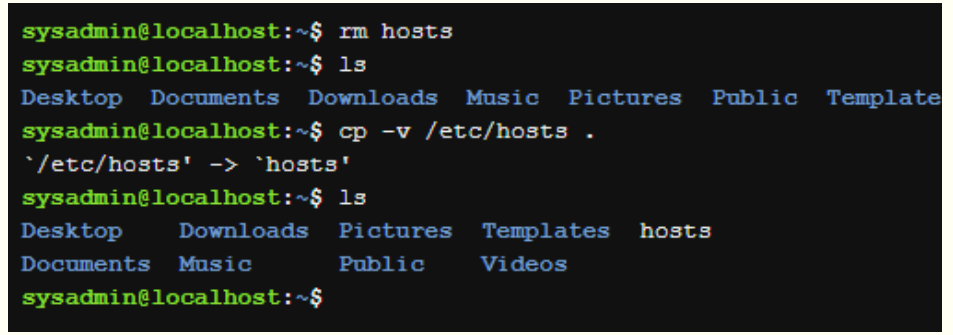
```
sysadmin@localhost:~$ ls -d /etc/[abcd]*
/etc/adduser.conf            /etc/blkid.conf            /etc/cron.weekly
/etc/adjtime                 /etc/blkid.tab             /etc/crontab
/etc/alternatives            /etc/ca-certificates       /etc/dbus-1
/etc/apparmor.d              /etc/ca-certificates.conf  /etc/debconf.conf
/etc/apt                     /etc/calendar              /etc/debian_version
/etc/bash.bashrc             /etc/cron.d                /etc/default
/etc/bash_completion.d       /etc/cron.daily            /etc/deluser.conf
/etc/bind                    /etc/cron.hourly           /etc/depmod.d
/etc/bindresvport.blacklist  /etc/cron.monthly          /etc/dpkg
sysadmin@localhost:~$
```

Files & Directories[copying]

- The syntax to copy files and directories using the cp command:
  - cp [OPTION] SOURCE DIRECTORY
- copy of the /etc/hosts file and place it in the current directory. Then list the contents of the current directory before and after the copy.
- you will remove the file, then copy it again, but have the system tell you what is being done. This can be achieved using the -v or --verbose option. Enter the following commands.
- copy the /etc/hosts file, using the period . character to indicate the current directory as the target

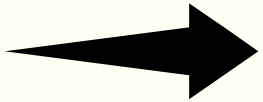






Files & Directories[copying]

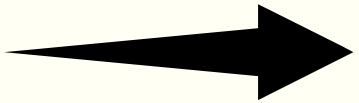
- copy from the source directory and preserve file attributes by using the -p option.



```
sysadmin@localhost:~$ rm hosts
sysadmin@localhost:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Template
sysadmin@localhost:~$ cd /etc
sysadmin@localhost:/etc$ ls -l hosts
-rw-r--r-- 1 root root 150 Jan 22 15:18 hosts
sysadmin@localhost:/etc$ cp -p hosts /home/sysadmin
sysadmin@localhost:/etc$ cd
sysadmin@localhost:~$ ls -l hosts
-rw-r--r-- 1 sysadmin sysadmin 150 Jan 22 15:18 hosts
sysadmin@localhost:~$
```

- Notice
  - that the date and permission modes were preserved.
  - that the timestamp in the output above is the same for both the original and the copy (Jan 22 15:18) in the example provided above. Your output may vary.

- copy using a different target name



```
sysadmin@localhost:~$ rm hosts
sysadmin@localhost:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Template
sysadmin@localhost:~$ cp -p /etc/hosts ~
sysadmin@localhost:~$ cp hosts newname
sysadmin@localhost:~$ ls -l hosts newname
-rw-r--r-- 1 sysadmin sysadmin 150 Jan 22 15:18 hosts
-rw-r--r-- 1 sysadmin sysadmin 150 Jan 22 16:29 newname
sysadmin@localhost:~$ rm hosts newname
sysadmin@localhost:~$
```

Files & Directories[copying]

- copy all files in a directory use the -R option. For this task, you will copy the /etc/udev directory and display the contents of the copied directory.



```
sysadmin@localhost:~$ mkdir Myetc
sysadmin@localhost:~$ cp -R /etc/udev Myetc
sysadmin@localhost:~$ ls -l Myetc
total 0
drwxr-xr-x 1 sysadmin sysadmin 32 Jan 22 16:35 udev
sysadmin@localhost:~$ ls -lR Myetc
Myetc:
total 0
drwxr-xr-x 1 sysadmin sysadmin 32 Jan 22 16:35 udev

Myetc/udev:
total 4
drwxr-xr-x 1 sysadmin sysadmin  56 Jan 22 16:35 rules.d
-rw-r--r-- 1 sysadmin sysadmin 218 Jan 22 16:35 udev.conf

Myetc/udev/rules.d:
total 8
-rw-r--r-- 1 sysadmin sysadmin  306 Jan 22 16:35 70-persistent-c
-rw-r--r-- 1 sysadmin sysadmin 1157 Jan 22 16:35 README
sysadmin@localhost:~$
```

Files & Directories[Moving]

- to remove a directory use the -r option to the rm command
- Note
  - that the rmdir command can also be used to delete directories, but only if the directory is empty (if it contains no files).
  - the -r option. This option removes directories and their contents recursively.

```
sysadmin@localhost:~$ touch premove
sysadmin@localhost:~$ ls
Desktop    Downloads  Pictures  Templates  premove
Documents  Music      Public    Videos
sysadmin@localhost:~$ mv premove postmove
sysadmin@localhost:~$ ls
Desktop    Downloads  Pictures  Templates  postmove
Documents  Music      Public    Videos
sysadmin@localhost:~$
```

Files & Directories[removing]

- Moving a file is analogous to a "cut and paste".
- The file is “cut” (removed) from the original location and “pasted” to the specified destination.
- Move a file in the local directory.

```
sysadmin@localhost:~$ ls
Desktop    Downloads  Myetc     Public     Videos
Documents  Music      Pictures  Templates
sysadmin@localhost:~$ rm -r Myetc
sysadmin@localhost:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Template
sysadmin@localhost:~$
```

Linux Command	Description
`touch premove`	Creates an empty file called `premove`
`mv premove postmove`	This command “cuts” the `premove` file and “pastes” it to a file called `postmove`
`rm postmove`	Removes `postmove` file


Thank
you